

Comparison of Background Radiation (Gamma Dose Rate) Between Mawlana Bhashani Science and Technology University (MBSTU) and Pabna University of Science and Technology (PUST)

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Session: 2020 -2021

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A Project submitted to the Department of Physics, Mawlana Bhashani Science and Technology University, as a partial fulfillment of the requirement for the degree of Bachelor of Science (Hons.) in Physics.



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June 2026

APPROVAL

A project entitled "**Comparison of Background Radiation (Gamma Dose Rate) Between Mawlana Bhashani Science and Technology University (MBSTU) and Pabna University of Science and Technology (PUST)**" is submitted by **Sabbir Ahmed Nirjon** (Student ID:PH-21016) as a partial fulfilment of the requirements for the degree of Bachelor of Science in Physics to the Department of Physics of Mawlana Bhashani Science and Technology University.

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ACKNOWLEDGEMENT

First and foremost, we would like to express our gratitude to Almighty ALLAH for providing the ability to complete the project work. We would like to express our heartfelt appreciation to our supervisor and mentor, Dr. Md. Moniruzzaman, Professor, Department of Physics, Mawlana Bhashani Science and Technology University, for his valuable suggestions, guidance, and strong assurance during the report's preparation. His knowledgeable discussion and helpful mind have been a great source of inspiration for us. We would like to express our heartfelt gratitude to all our honorable teachers for their inspiration and a keen interest in this project. And last, we would like to thank our family and everyone close to us while doing project work as well as all aspects of our lives.

Sabbir Ahmed Nirjon

June 2026

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ABSTRACT

This study compares the ambient gamma dose rates, annual effective doses (AED), and excess lifetime cancer risks (ELCR) at two public university campuses in Bangladesh: Mawlana Bhashani Science and Technology University (MBSTU) in Tangail and Pabna University of Science and Technology (PUST) in Pabna. We selected 25 monitoring points at each campus to cover the radiation environment completely. At MBSTU, the average gamma dose rates ranged from 0.077 $\mu\text{Sv/hr}$ to 0.232 $\mu\text{Sv/hr}$. The highest levels were recorded at the Medical Center (Inside) and JAMH Gate. At PUST, the average gamma dose rates ranged from 0.120 $\mu\text{Sv/hr}$ to 0.254 $\mu\text{Sv/hr}$, with peak levels noted at the Convention Center and Medical Center. The annual effective doses varied from 0.094 mSv to 0.285 mSv for MBSTU and from 0.147 mSv to 0.312 mSv for PUST. The excess lifetime cancer risk values ranged from 0.331×10^{-3} to 0.996×10^{-3} for MBSTU and from 0.515×10^{-3} to 1.090×10^{-3} for PUST. Overall, PUST showed slightly higher gamma dose rates at most monitoring points compared to MBSTU. Importantly, PUST is near the Rooppur Nuclear Power Plant, so radiation monitoring is vital for future environmental and radiological studies. All measured values at both campuses were within the safety limits set by the United Nations Scientific Committee on the Effects of Atomic Radiation (UNSCEAR) and the International Commission on Radiological Protection (ICRP). This confirms that both university environments are radiologically safe for students, faculty, staff, and the public.

Chapter 1

1.1 Introduction

Ionizing radiation has always been part of the natural world since the Earth formed. Every living organism on this planet faces exposure to it every day. The sources of this background radiation are diverse. They include cosmic rays from outer space, naturally occurring radioactive materials (NORMs) such as uranium, thorium, and potassium-40 found in the soil and rocks, and radionuclides that enter our bodies through the food we consume, the water we drink, and the air we breathe. In addition to these natural resources, human activities have introduced artificial sources. These include medical imaging equipment, industrial radiography, research reactors, and nuclear power plants. The United Nations Scientific Committee on the Effects of Atomic Radiation reports that the global average annual effective dose from natural background radiation is about 2.4 mSv [1]. This figure can vary greatly from region to region due to local geology, altitude, and land use. This variation highlights the importance of monitoring radiation levels at specific sites. In Bangladesh, increasing research on environmental radiation shows a growing awareness of the need for location-specific baseline data. A nationwide spatial analysis, based on decades of published and unpublished studies, found that absorbed dose rates across the country vary widely. They range from as low as 0.02 nGy/h to as high as 745.86 nGy/h, with a national average of 64.16 nGy/h. This average is close to the world average of 59 nGy/h [2]. Significant differences in radiation levels were noted, mostly due to variations in geology and soil composition. This means that even two nearby districts can have significantly different background radiation levels. Thus, generalizing from one location to another without actual measurement can pose risks from a public health standpoint.

This study focuses on two public universities as measurement sites: Pabna University of Science and Technology (PUST) in Pabna district and Mawlana Bhashani Science and Technology University (MBSTU) in Tangail district. Both are science and technology universities with large student populations living and studying on campus. They are located in the floodplain-dominated central region of Bangladesh. However, they sit in different districts with possibly different geology and land-use patterns, which are known to affect ambient gamma dose rates. Comparing the two sites using

systematic measurements allows us to see how the radiological environment varies between similar university campuses in different parts of the country. It is important to note that PUST in Pabna is about 31 km from the Rooppur Nuclear Power Plant (RNPP) at Ishwardi. Pre-operational surveys done in the 30-km zone around the RNPP [3] found ambient dose equivalent rates ranging from 0.11 to 0.15 $\mu\text{Sv/h}$, which were attributed to natural background radiation. In one of the earliest baseline studies at the proposed RNPP site, measured outdoor gamma dose rates between 0.12 and 0.23 $\mu\text{Gy/h}$, with an average of 0.17 $\mu\text{Gy/h}$ [4]. This confirmed that the area's pre-operational radiological environment was within normal limits. While PUST is beyond the formal 30-km monitoring zone, recording baseline gamma dose rates at the campus level now, as the plant enters its commissioning phase, is valuable for any future post-operational comparisons.

Several studies conducted at university and hospital campuses across Bangladesh provide useful comparisons for understanding measurements at PUST and MBSTU. However, in other studies the outdoor gamma dose rates measured in Dhaka University campus, which ranged from 0.067 to 0.19 $\mu\text{Sv/h}$, with an average of 0.138 $\mu\text{Sv/h}$ [5]. This translates to a mean annual effective outdoor dose of 0.242 mSv/y, well within the ICRP-recommended public exposure limit of 1 mSv/y [6]. Researchers from MBSTU's own Department of Physics carried out a gamma radiation monitoring study in the New Market Thana area of Dhaka. They found dose rates between 0.103 and 0.168 $\mu\text{Sv/h}$, with an average of 0.135 $\mu\text{Sv/h}$ and an annual effective dose of 0.238 mSv/y [7]. Studies at hospital campuses, where medical radiation sources add to the natural background, recorded somewhat higher dose rates. For example, measurements at BSMMU in Dhaka showed an average outdoor dose rate of about 0.211 $\mu\text{Sv/h}$ and an average annual effective dose of 0.368 mSv/y [8]. These figures help clarify what might be expected at a typical Bangladeshi campus environment without nearby medical or nuclear installations.

In addition to dose rate, researchers often calculate the excess lifetime cancer risk (ELCR) related to long-term gamma exposure as a health indicator. It was assessed that at indoor and outdoor gamma radiation at the University of Port Harcourt Teaching Hospital and found average annual effective doses of 0.92 mSv/y (indoor) and 0.31

mSv/y (outdoor) [9]. Their ELCR values exceeded the global reference threshold of 0.29×10^{-3} . This shows that even university and hospital environments without nuclear facilities can have higher radiological levels if building materials or local geology are enriched. In Bangladesh, ELCR values from campus monitoring studies have generally stayed within internationally accepted limits. However, they vary enough between locations to warrant site-specific assessments instead of assuming uniformity across institutions [8,10]. To perform comparative in-situ radiological survey of terrestrial background gamma dose rates on the campuses of PUST and MBSTU. Portable radiation detecting devices will be employed for the purpose following recognized protocols of conducting surveys. Absorbed dose rate, annual effective dose equivalent and excess lifetime cancer risk will be computed based on the measured dose rate at both campuses. The result will reveal whether ambient gamma dose rates differ significantly from each other at the two campuses or not. It will also be known whether they adhere to internationally acceptable limits or not. Special focus will be made to identify if PUST which is located in proximity to recently commissioned nuclear power plant reflects any perceptible departure from natural background level or not. Result will enhance a very limited database of campus-based monitoring of radiation doses all over Bangladesh.

1.1.2 Aims and Objectives

Aim

The study's objective is to determine and compare the ambient gamma radiation levels (background dose rate) at Pabna University of Science and Technology (PUST) and Mawlana Bhashani Science and Technology University (MBSTU), as well as assess related radiological hazard parameters and potential impacts on students', staff's and neighbors' health.

Specific Objectives

1. To measure ambient gamma dose rates at selected outdoor locations within the campuses of PUST and MBSTU using calibrated portable radiation detection instruments.

2. To evaluate and compare background radiation levels between the two campuses, identifying any statistically meaningful differences between the Pabna and Tangail settings.
3. To calculate radiological hazard parameters for both sites, including:
 - absorbed dose rate
 - annual effective dose equivalent
 - excess lifetime cancer risk (ELCR)
4. To assess whether radiation levels at PUST, situated approximately 31 km from the Rooppur Nuclear Power Plant, show any detectable deviation from expected natural background compared to MBSTU.
5. To determine whether ambient gamma dose rates at both campuses remain within internationally accepted safety limits as defined by ICRP and endorsed by the Bangladesh Nuclear Safety and Radiation Control (NSRC) framework.
6. To provide a radiological baseline dataset for both institutions that can support future environmental monitoring and be referenced in any post-operational assessment of the RNPP's broader radiological influence.

1.1.3 Significance of the Study

Routine campus wise background gamma-ray survey measurements are rarely seen in Bangladesh universities. Till now, there is no published in-situ gamma-ray baseline measurement data for PUST nor MBSTU campus areas which this study aims to fulfill. Other than the academic interest this work may attract, there are some practical significances of this data:

- It presents students, faculties and locals of both universities with fact-based knowledge of their everyday surrounding radiation conditions.
- It adds valuable information to our countrywide radiological monitoring dataset just as RNPP has initiated Bangladesh's nuclear energy era.
- They establish baseline levels we can refer to in any future epidemiological research, especially longterm research into radiation exposure of university students nationwide in Bangladesh.

- They allow us to identify background levels vs. any potential contribution from flora at PUST. This information can be useful to regulatory agencies AND the public when explaining what is 'normal' in this area.

On a larger scale, this type of study shows that radiological monitoring can be done at the campus level. It's not just limited to industrial plants or nuclear facilities. Schools should be subject to the same degree of scrutiny since many people are there for long periods of time.

1.1.4 Layout of the Project

This project is configured with into seven chapters: Introduction, Theoretical Context, Materials and Methods, Radiation and Human Health, Results and Discussion, and Conclusions.

Chapter 2

Theoretical Context

2.1 Radiation

Radiation refers to the emission and transfer of energy through space or a material medium in the form of waves or particles. It is a fundamental physical phenomenon that plays a vital role in many natural and human-made processes. Radiation can be broadly classified into two categories:

- (a) wave-like radiation, which includes microwaves, X-rays, gamma (γ) rays, and other forms of electromagnetic radiation, and
- (b) particle-like radiation, such as alpha (α) and beta (β) particles. All living organisms are continuously exposed to background radiation, a natural and ever-present part of the environment that originates from a variety of natural sources.

2.2 Sources of Background Radiation

There has been background radiation in the cosmos since its inception. Sources of background radiation mostly belong to two groups. These consist of

- (a) natural sources and
- (b) artificial sources.

2.2.1 Natural Sources

Background radiation, also referred to as ionizing radiation, is released by a variety of natural sources. Air, water, food, minerals, radioactive substances, and cosmic background radiation are a few examples.

Cosmic Radiation

Like a constant flow of rain, cosmic radiation is constantly sent to Earth by the sun and stars. Our exposure to cosmic radiation can differ based on factors including air quality, elevation, and the Earth's magnetic [11].

Terrestrial Radiation

Terrestrial radiation comes from the Earth. Radium, thorium, and uranium are naturally occurring radioactive elements found in rock and soil. Radon is the most prevalent natural form of background radiation exposure to which Americans are subjected. In fact, there is always some radon in the air we breathe. Additionally, all plants and animals contain radioactive carbon and potassium, and all water contains small amounts of dissolved uranium and thorium. Some substances, such as radon gas, are breathed, while others such as certain foods and drinks are ingested. Americans receive varying amounts from terrestrial sources depending on where in the world they live, but on average places with higher concentrations of uranium and thorium in their soil receive higher doses [11].

Internal Radiation

Everyone is a source of radiation exposure because of the radioactive potassium-40 and carbon-14 that are always present in their body. The variance in dosage across individuals is not as significant as that linked to cosmic and terrestrial sources [12].

2.2.2 Artificial Sources

There is also a risk of ionizing radiation originating from human-made sources, which often occurs during medical treatments. The doses from diagnostic X-rays are generally much lower than the ambient radiation when considering dose effectiveness. On the other hand, radiation therapy doses are usually confined to the affected tissues and can exceed ambient background radiation by significant margins. A small amount of man-made background radiation, unrelated to medical applications, is emitted by common products, facilities that use radioactive materials, educational and research institutions, nuclear power plants, and associated structures like fuel preparation plants and uranium mills, in addition to federal sites that produce nuclear weapons. Additionally, tobacco smoke contains radio nuclides that further expose smokers to radiation [13].

Medical Procedures

Radiation is used for several purposes in medicine. Probably the most famous example would be the X-ray machine. X-ray machines use radiation to detect broken bones and

find diseases. These machines are regulated by Health. Nuclear medicine also uses radioactive isotopes to detect, and in some cases treat diseases like cancer [14]. Millirem or mrem is a unit of radiation dose.

Atmospheric testing

Fallout is radioactive material emitted into the atmosphere from nuclear explosions that fall back onto Earth. Nuclear weapons tests above ground between the end of World War II and as late as 19 October 1980 deposited radioactive fallout into the atmosphere. When the fallout came down out of the sky it became lodged in everything on Earth. Most of these nuclides had short half-lives and have since decayed away but others persist to the present day. We still receive small doses from them each year that become less every year [14].

Nuclear fuel cycle

Nuclear power plants (NPPs) use uranium to drive a chain reaction that produces steam, which in turn drives turbines to produce electricity. As part of their normal activities, NPPs release regulated levels of radioactive material which can expose people to low doses of radiation. Similarly, uranium mines, fuel fabrication plants and radioactive waste facilities release some radioactivity that contributes to the dose of the public.

2.3 Types of radiation

Radiation can be divided into two categories based on its impact on health.

- i. Ionizing radiation
- ii. non-ionizing radiation.

2.3.1 Ionizing radiation

UV radiation of higher frequency has enough energy to break molecular bonds. X-rays and gamma rays, at the far end of the spectrum, have wavelengths of 100 millionths of a meter (or shorter) and frequencies up to a hundred billion Hertz. This region is where the energy of radiation is very high. Free radicals are produced in objects when exposed to ionizing radiation even at ambient temperatures or lower. This generation of free radicals is caused by ionization radiation and differs greatly from the free radical production by (low-temperature) heating. Even relatively small doses of ionizing

radiation (that is, smaller than those that would cause significant heating) can have a special risk to living organisms because of the production of free radicals. Free radicals and ionizing radiation can cause direct damage either by breaking the DNA molecule or by ionizing the DNA, respectively. Radioactive substances, Xray machines, and particle accelerators are just some of the places we find radioactive emissions in our world. This invisible phenomenon can be indirectly detected by the human body using the Geiger counter. In some materials, ionization radiation sometimes produces secondary radiation of visible light (radioluminescence, Cherenkov radiation). It has many applications that benefit people from fields such as construction, research and medicine, but when used improperly and in the wrong way it is a danger to health [15].

2.3.2 Nonionizing radiation

Non-ionizing radiation sources are used and exposed to by people on a daily basis. The energy of this radiation is not sufficient to ionize atoms or molecules. Non-ionizing radiation is used in microwave ovens, global positioning systems, cellular telephones, television stations, FM and AM radio, baby monitors, cordless phones, garage-door openers and ham radios. Other forms include the earth's magnetic field and magnetic field exposure due to the presence of transmission lines, domestic wiring and electrical appliances. These are known as extremely low frequency (ELF) waves. It is most clearly seen in a microwave oven where the radiation is used to increase the vibration speed of water molecules, thus generating heat [15].

2.4 Radioactive decay

In an atom's nucleus, the result of nuclear instability is the phenomenon of radioactivity. It is called the Becquerel effect and was discovered by Henry Becquerel in 1896. When an atom's nucleus is unstable, it emits energy by the process of radioactivity. A small amount of uranium compound was placed on top of the photographic plates and black paper was placed over it. Further examination of these plates indicated that these plates were exposed. This event has been dubbed "radioactive decay. Radioactive elements are those that are radioactive and emit radiation [16].

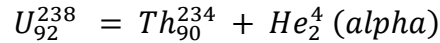
2.4.1 Types of radioactive decay

There are three categories of radioactive decay, namely

- a) Alpha decay
- b) Beta decay
- c) Gamma decay and
- d) Neutron decay.

Alpha decay

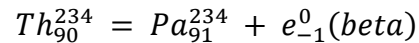
Alpha decay is the process by which a proton from an alpha particle releases its nucleus. It is assumed that the extremely stable alpha particle is the helium nucleus. It consists of a pair of protons and two neutrons. The following is an illustration of uranium-238's alpha decay. Alpha Decay Example:



The process by which isotopes change into a component of a stable nucleus is known as transmutation [16].

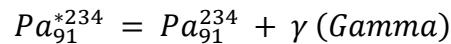
Beta Decay

A beta particle is commonly referred to as an electron, but it can also be a positron. If the reaction involves electrons, the nucleus releases neutrons one at a time, increasing the number of protons. The process of beta decay is demonstrated below:



Gamma Decay

Orbiting electrons in the nucleus possess energy, and when an electron transitions from a high-energy level to a low-energy level, a photon is emitted. Similarly, in the nucleus, a high-energy photon called a gamma ray is released when it rearranges into a lower energy state [16].



Neutron Decay

Neutron radiation occurs when neutrons are ejected from the nucleus by nuclear fission and other processes. The nuclear chain reaction is an example of nuclear fission, where a neutron being ejected from one fissioned atom causes another atom to fission, ejecting more neutrons. Unlike other radiation, neutron radiation is absorbed by materials with lots of hydrogen atoms, like paraffin wax and plastics.

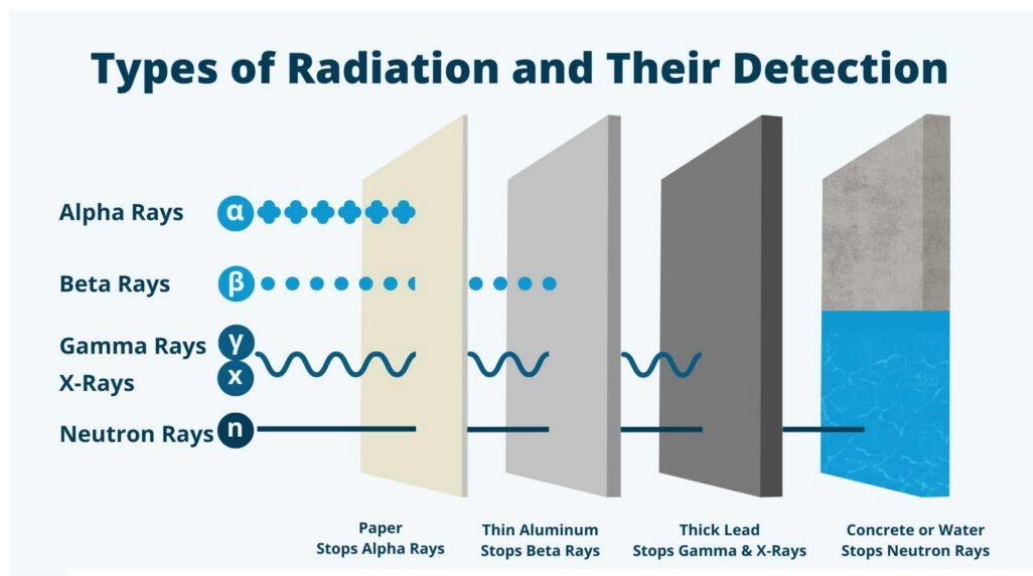


Figure 1: Radioactive decay types [14]

Chapter 3

Materials and Methods

3.1 Analytical Techniques

- I. GAMMA SCOUT Geiger counter
- II. GPS survey instrument
- III. Microsoft Excel

The current investigation was carried out in March 2025. Inside the MBSTU campus, gamma radiation was measured because it was thought that the radiation from the campus's numerous nuclear technologies would be harmful to the students, teachers, and other individuals. Gamma-Scout detector measurements were carried out in situ with a portable device. The detector was positioned one meter above the ground on a tripod table.



Figure 2: In-situ radiation dose rate monitoring set up inside the MBSTU campus.

3.2 Gamma Scout Geiger Counter

Gamma-Scout detectors are used to identify radioactive samples, measure personal exposure, track ambient radiation levels, and locate the sources of radiation leakages.

The detectors help in determining:

- ✓ Cumulative exposure function I (dosimeter-like function)
- ✓ A visual signal when the annual dosage threshold is crossed
- ✓ Support for several units ($\mu\text{Sv/hr}$, mRem/hr)
- ✓ The pulse mode (total count or rate)
- ✓ USB data downloads to a computer
- ✓ Programmable alert feature.

3.2.1 Calibration

There will be no environmental radiation to wear down the counter tube, so it will not have to be re-calibrated. GAMMASCOUT rapidly, reliably and permanently reports the input of radioactivity. How pulse/ min is converted to a dosage rate depends on the amount of pulse input. We are using four different ratios to approximate this nonlinear

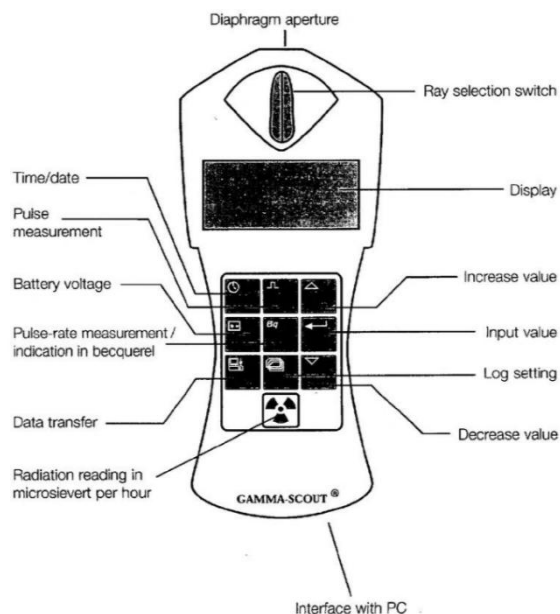


Figure 3: Gamma Scout Geiger counter is used for real-time environmental Radiation measurement.

relationship. The unit is 142 pulses per minute at 100 percent input (approximately 0.2 Sv/hr).

GPS survey Instrument

GPS instruments were used to determine the location of the radiation dose rate precisely.



Figure 4: GPS survey instrument.

Microsoft Excel

I used Microsoft Excel to analyze the observing data and to make charts.

3.3 Radiometric Analysis

3.3.1 Gamma Dose Rate (GDR)

The amount of energy per unit time per unit mass of an absorbing medium is known as the absorbed dose rate. The units of absorbed dose rate are (a) micro-sievert per hour ($\mu\text{Sv/hr}$) or (b) milli-rem per hour (mRem/hr). The gamma gives the gamma dose rate in $\mu\text{Sv/hr}$ directly.

3.3.2 Annual Effective Dose (AED)

The amount of radiation that is absorbed by the whole body in a year is called the annual effective dose. It is calculated from the equation as [7,17].

$$\text{AED (mSv)} = [\text{GDR } (\mu\text{Sv/hr})/1000] \times 8760 \times \text{OF} \quad (1)$$

Where OF is the outdoor occupancy factor having a value of 0.2 [18].

3.3.3 Excess Lifetime Cancer Risk (ELCR)

Potential carcinogenic effects that are characterized by estimating the probability of cancer incidence in a populace of individuals for a particular lifetime from projected exposures and intakes is called ELCR. It is assessed from the following equation [17].

$$\text{ELCR} = \text{AED} \times \text{RF} \times \text{DL} \quad (2)$$

Where RF is the risk factor per Sievert, the value of RF is 0.057 Sv^{-1} [17], and DL is the Duration of Life of Bangladeshi people which is 73.82 years (2022)

3.3.4 Organ Annual Effective Dose (OAED)

The annual effective dose is absorbed by an individual organ of the body like the Liver, Kidney, Testes, Bone Marrow, Ovaries, etc. is called organ annual effective dose. It is calculated from the equation [6],

$$\text{OAED} = \text{AED} \times \text{CR} \quad (3)$$

Where CR is the reduction coefficient, the CR value for the Kidney is 0.62 [19].

3.3.5 Organ Cancer Risk (OCR)

The estimated likelihood that ionizing radiation exposure will cause cancer in a particular organ or tissue over a person's lifetime is called OCR. Cancer risk for an individual organ of the body assessed from the equation [19].

$$\text{OCR} = \text{OAED} \times \text{RF} \times \text{DL} \quad (4)$$

Chapter 4

Radiation and Human Health

4.1 Health Effects of Ionizing Radiation

Radiation damage to tissue and/or organs depends on the dose of radiation received, or the absorbed dose. Beyond certain thresholds, radiation can impair the functioning of tissues and/or organs and can produce acute effects such as skin redness, hair loss, radiation burns, or acute radiation syndrome. These effects are more severe at higher doses and higher dose rates. For instance, the dose threshold for acute radiation syndrome is about 1 Sv (1000 mSv). Therefore, dentists must carefully select their patients and only perform radiographs when there is a strong justification that the radiograph will yield valuable information for diagnosis or treatment [20]. Epidemiological studies on populations exposed to radiation, such as the survivors of the atomic bombings or radiotherapy patients, showed a significant increase of cancer risk at doses above 100 mSv. More recently, some epidemiological studies in individuals exposed to medical exposure during childhood (pediatric CT) have suggested that cancer risk may increase even at lower doses (between 50-100 mSv).

4.1.1 Radiation Damage

Ionizing radiation harms the genetic material in reproductive cells, leading to mutations that can be passed down through generations. The mutagenic effects of radiation were first identified in the 1920s, and since then, it has served as a key tool in genetic research for generating new mutations in experimental organisms. While there has always been concern about occupational exposure to high radiation levels, it wasn't until during and after World War II that significant efforts were made to assess the genetic impacts of radiation on whole populations. This movement was driven by worries regarding the effects of very large radiation sources being developed in the nuclear sector, radioactive fallout from atmospheric nuclear tests, and the growing utilization of radiation in medical diagnosis and treatment. In 1956, the National Academy of Sciences-National Research Council (NAS-NRC) formed the Committee on the Biological Effects of Atomic Radiation (known as the BEAR Committee), which paved the way for later NAS-NRC committees studying Biological

Effects of Ionizing Radiation (the BEIR committees, including this BEIR V report). Additionally, a series of reports from the U.N. Scientific Committee on the Effects of Atomic Radiation (UNSCEAR) have also explored the genetic consequences of radiation exposure on populations. Dental radiation exposure has the potential to leukemia and cancer [21,22].

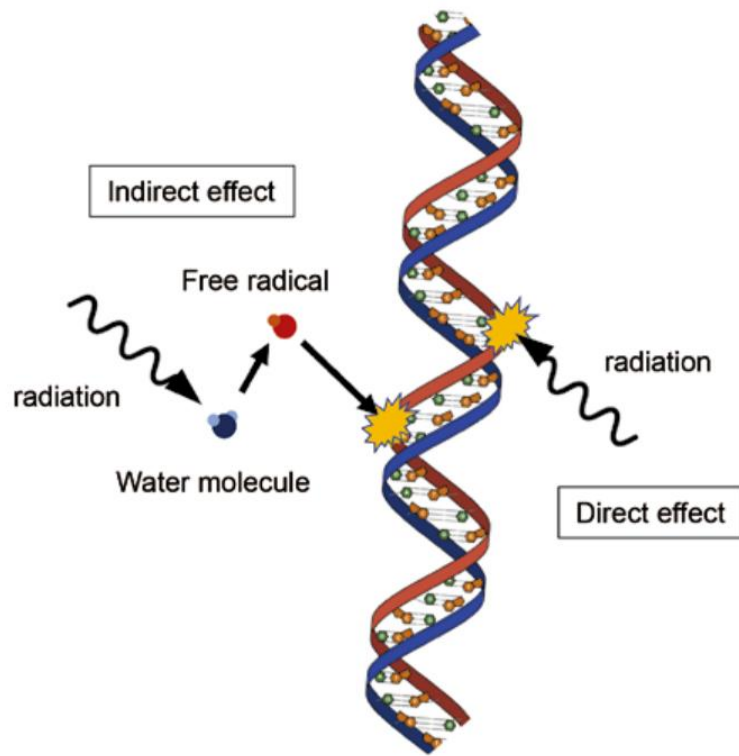


Figure 5: Impact of radiation on DNA.

4.1.2 Radiation and Cancer

Estimated risk of cancer due to radiation exposure in diagnostic imaging has been extrapolated from studies of people exposed to very high radiation doses (eg, survivors of the atomic bomb explosions at Hiroshima and Nagasaki). Direct epidemiological evidence from human populations demonstrates that exposure to ionizing radiation increases the risk of some cancers when doses exceed approximately 50 to 100 mSv for protracted exposure (eg, in occupational settings) or 10 to 50 mSv for acute exposure (eg, from atomic bomb exposure) [23]. Risk is higher in young patients because

- They live longer, giving cancer more time to develop.
- More cellular growth (and thus susceptibility to DNA damage) occurs in the young.

For a 1-year-old who has a CT scan of the abdomen, estimated lifetime risk of developing cancer is increased by 0.18% [14]. If an older patient has this test, risk is

lower. Risk also depends on the tissue being irradiated. Lymphoid tissue, bone marrow, blood, and the testes, ovaries, and intestines are considered very radiosensitive; in adults, the central nervous and musculoskeletal systems are relatively radioresistant.

4.1.3 Kidney Exposure

The kidneys play an important role in the production of erythropoietin for red blood cell production, filtration of wastes/electrolytes from the blood, and modulation of blood pressure through fluid/electrolyte balance [24]. The kidneys are highly radiosensitive, resulting in a limited dose of radiation therapy (RT) that can be safely delivered to kidney-containing treatment volumes in pelvis RT [25].

Radiotherapy with or without chemotherapy for pelvic malignancies such as gastrointestinal cancers, gynecologic cancers, lymphomas, and sarcomas of the upper abdomen and during total body irradiation (TBI) may result in radiation-induced kidney injury, especially radiation nephropathy (RN) [24,26,27]. The use of TBI for preparation of bone marrow transplantation (BMT) and the effects of radio nuclide therapies have led to an increase in the incidence of clinical RN. BMT nephropathy typically progresses over many years and is characterized by proteinuria, high blood pressure and an inability to concentrate on the urine [27,28].

However, because of a threshold dose, most or all structures of the kidney are damaged if it receives enough ionizing radiation. First step is glomerular injury (involving at least the endothelium and mesangium) and then progression to glomerular scarring from thrombotic microangiopathy (TMA). The expression of tubular injury seems to be somewhat delayed but can be triggered at the same time as the glomerular injury. Oxidative injury to the glomeruli could play a mechanistic role. Tubules which have been denuded might permit interstitial access from mediators that are able to escape from damaged glomeruli. Local mediator expression, such as TGF β 1 or activation of renin-angiotensin system could be key in creating tubulointerstitial scarring.

Chapter 5

Results and Discussions

5.1 Mawlana Bhashani Science and Technology University Campus

Area:

5.1.1 Results:

Table 1: Mean Gamma Dose Rate, Annual Effective Dose and Excess Lifetime Cancer Risk in different locations of MBSTU campus.

Serial No.	Location Name	Latitude	Longitude	Mean Gamma Dose Rate ($\mu\text{Sv/hr}$)	Annual Effective Dose (mSv)	Excess Lifetime Cancer Risk $\times 10^{-3}$
1	1 st Gate MBSTU	24°14'01.5''N	89°53'32.8''E	0.077	0.094	0.331
2	Shadhinota Chatter MBSTU	24°14'05.7''N	89°53'32.2''E	0.079	0.097	0.339
3	Old Administrative Building	24°14'06.5''N	89°53'31.1''E	0.091	0.112	0.391
4	Administrative Building MBSTU	24°14'08.0''N	89°53'27.6''E	0.099	0.121	0.425
5	Mukta Mancha MBSTU	24°14'07.1''N	89°53'25.2''E	0.135	0.166	0.579
6	Medical Center (Front)	24°14'08.9''N	89°53'25.6''E	0.207	0.254	0.889
7	Medical Center (Inside)	24°14'09.1''N	89°53'26.2''E	0.232	0.285	0.996

8	Cafeteria	24°14'12.8''N	89°53'27.0''E	0.116	0.142	0.498
9	Academic Building 1	24°14'15.3''N	89°53'26.5''E	0.140	0.172	0.601
10	Teacher's Dormitory Gate	24°14'19.0''N	89°53'26.6''E	0.159	0.195	0.682
11	Hall Chatter	24°14'19.9''N	89°53'27.1''E	0.152	0.186	0.652
12	JAMH, Gate	24°14'20.1''N	89°53'29.3''E	0.196	0.240	0.841
13	Academic Building 3	24°14'17.2''N	89°53'29.1''E	0.169	0.207	0.725
14	Academic Building 2	24°14'15.4''N	89°53'31.3''E	0.126	0.155	0.541
15	Krishnochura Lane	24°14'12.9''N	89°53'32.1''E	0.123	0.151	0.528
16	Mannan Hall (Front)	24°14'19''N	89°53'28''E	0.144	0.177	0.618
17	Central Field	24°14'20''N	89°53'21''E	0.130	0.159	0.558
18	Research Building	24°14'16''N	89°53'28''E	0.113	0.139	0.485
19	Physical Exercise Building	24°14'12''N	89°53'27''E	0.165	0.202	0.708
20	Central Temple	24°14'10''N	89°53'31''E	0.131	0.161	0.562

21	Dean's Complex	24°14'06''N	89°53'29''E	0.159	0.195	0.682
22	Multipurpose Building (Back)	24°14'09''N	89°53'26''E	0.163	0.200	0.700
23	Central Mosque	24°14'10''N	89°53'23''E	0.153	0.188	0.657
24	5 th Gate	24°14'09''N	89°53'31''E	0.195	0.239	0.837
25	Beside Research Pond	24°14'13''N	89°53'29''E	0.122	0.150	0.524
Average				0.14372	0.17548	0.61396

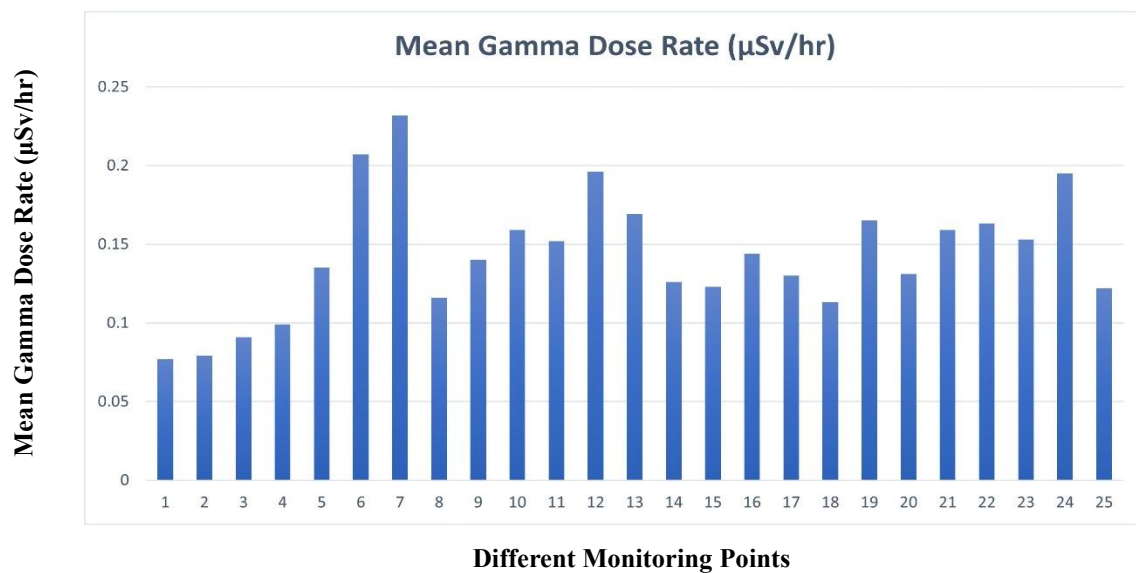


Figure 6: Variation of Mean Gamma Dose Rate in various place in MBSTU.

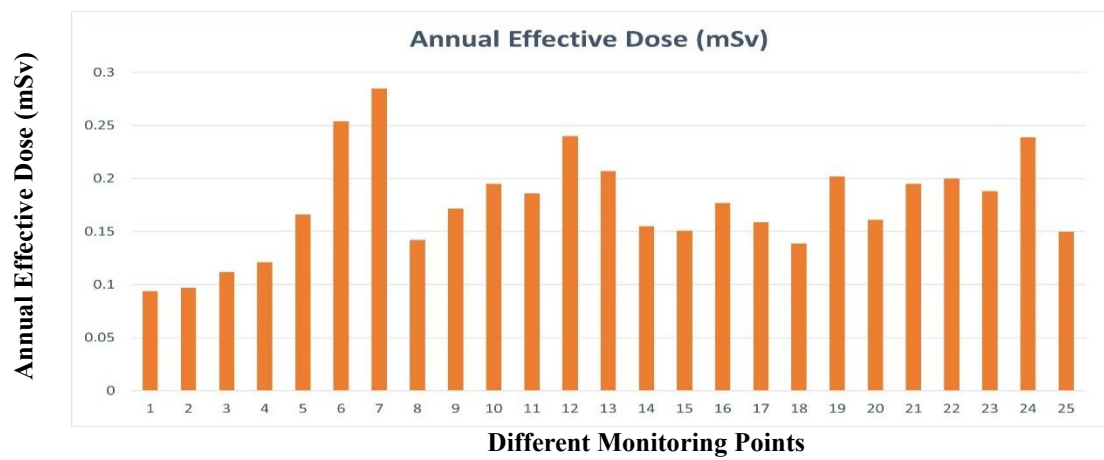


Figure 7 Variation of Annual Effective Dose Rate in various place in MBSTU.

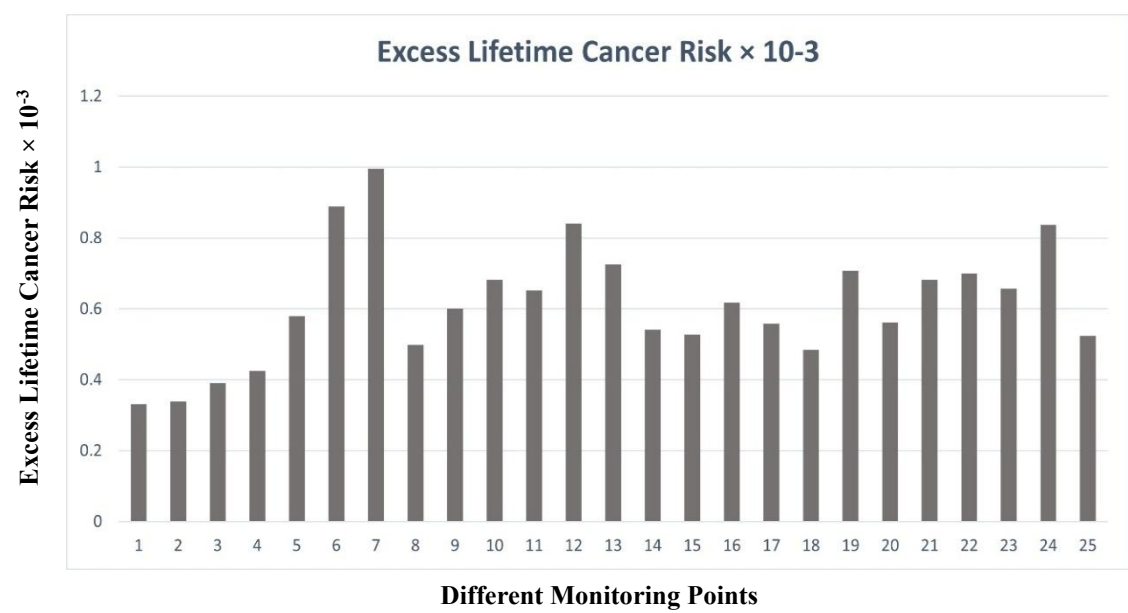


Figure 8: Variation of Excess Lifetime Cancer Risk in various place in MBSTU.

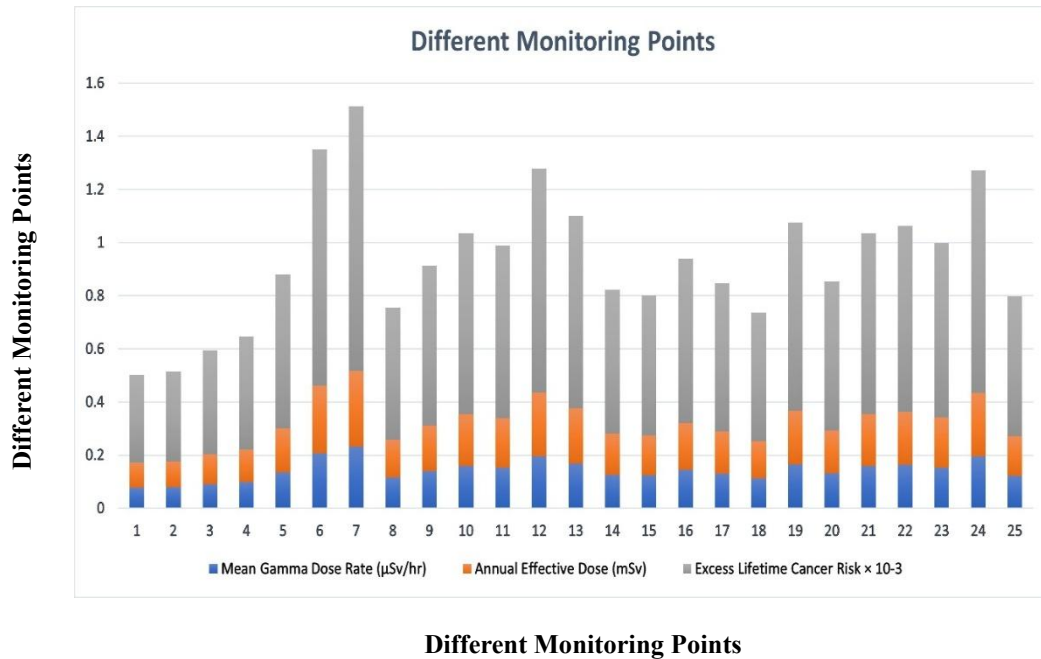


Figure 9: Variation of mean gamma dose rate, annual effective dose, and excess lifetime in MBSTU.

5.1.2 Discussion:

The variation of mean gamma dose rate, annual effective dose (AED), and excess lifetime cancer risk (ELCR) at different monitoring locations of the Mawlana Bhashani Science and Technology University (MBSTU) campus is presented in Figure 4. The measured mean gamma dose rates ranged from $0.077 \mu\text{Sv h}^{-1}$ at the 1st Gate MBSTU to $0.232 \mu\text{Sv h}^{-1}$ at the Medical Center (Inside), indicating noticeable spatial variation in the natural background radiation level across the campus. Similar variations in environmental gamma radiation have been reported in university campuses and urban environments worldwide, where local geology, radionuclide distribution, and anthropogenic structures significantly influence background radiation levels [29–32].

The comparatively higher gamma dose rates observed at the Medical Center (Inside), Medical Center (Front), JAMH Gate, and 5th Gate locations may be attributed to the presence of concrete structures, construction materials, and variations in soil composition. Building materials such as cement, brick, sand, and aggregates are known

to contain naturally occurring radionuclides including ^{238}U , ^{232}Th , and ^{40}K , which can contribute to elevated gamma radiation levels in localized environments [33,34]. In contrast, relatively lower dose rates recorded at the 1st Gate, Shadhinota Chatter and Old Administrative Building areas may reflect lower radionuclide concentrations in the surrounding soil and reduced influence of dense construction materials.

The calculated AED values varied from 0.094 to 0.285 mSv y^{-1} , while ELCR values ranged from 0.331×10^{-3} to 0.996×10^{-3} . Since both parameters are directly proportional to the measured gamma dose rate, locations exhibiting higher radiation levels also showed elevated AED and ELCR values. However, all estimated values remain within the range typically reported for natural background radiation environments and are comparable to those observed in similar environmental monitoring studies conducted in Bangladesh and other countries [30,31,35,36].

The worldwide average outdoor absorbed dose rate from terrestrial gamma radiation is reported by UNSCEAR to be approximately 59 nGy h^{-1} , corresponding to an annual effective dose of about 0.07 mSv y^{-1} , although significant regional variations may occur depending on geological and environmental factors [37]. The measured values at MBSTU are slightly higher than the global outdoor average but remain within the range commonly observed in naturally occurring radiation environments. Similar findings have been reported in environmental radiation surveys conducted in Dhaka, Chattogram, Cyprus, India, and other regions where local geology and construction materials influenced radiation levels without posing significant health hazards [30,32,35,38].

The estimated ELCR values indicate a very low probability of developing radiation-induced cancer due to environmental gamma exposure. These values are consistent with previous studies that concluded natural background radiation contributes only a minor fraction to overall lifetime cancer risk under normal environmental conditions [31,36,39]. Therefore, based on the measured gamma dose rates and derived radiological parameters, the MBSTU campus can be considered radiologically safe for students, faculty members, staff, and visitors. The results obtained also provide valuable baseline data for future environmental radiation monitoring and comparison studies within the region.

5.2 Pabna University Of Science and Technology Campus Area:

5.2.1 Results:

Table 2: Mean Gamma Dose Rate, Annual Effective Dose and Excess Lifetime Cancer Risk in different locations of PUST campus.

Serial No.	Location Name	Latitude	Longitude	Mean Gamma Dose Rate ($\mu\text{Sv/hr}$)	Annual Effective Dose (mSv)	Excess Lifetime Cancer Risk $\times 10^{-3}$
1	PUST Main Gate	24°14'01.5''N	89°53'32.8''E	0.120	0.147	0.515
2	PUST Central Mosque	24°14'05.7''N	89°53'32.2''E	0.156	0.191	0.670
3	PUST Central Field	24°14'06.5''N	89°53'31.1''E	0.131	0.161	0.562
4	PUST Shaheed Minar	24°14'08.0''N	89°53'27.6''E	0.160	0.196	0.687
5	PUST Shadhinot a Chatter	24°14'07.1''N	89°53'25.2''E	0.188	0.231	0.807
6	PUST Cafeteria	24°14'08.9''N	89°53'25.6''E	0.205	0.251	0.880
7	PUST TSC	24°14'09.1''N	89°53'26.2''E	0.149	0.183	0.640

8	PUST Central Library	24°14'12.8''N	89°53'27.0''E	0.178	0.218	0.764
9	PUST Medical Center	24°14'15.3''N	89°53'26.5''E	0.247	0.303	1.060
10	PUST 2 nd Administr ative Building	24°14'19.0''N	89°53'26.6''E	0.204	0.250	0.876
11	PUST Power Station	24°14'19.9''N	89°53'27.1''E	0.145	0.178	0.622
12	Residence of VC, PUST	24°14'20.1''N	89°53'29.3''E	0.209	0.256	0.897
13	PUST Conventio n Center	24°14'17.2''N	89°53'29.1''E	0.254	0.312	1.090
14	PUST Central Temple	24°14'15.4''N	89°53'31.3''E	0.224	0.275	0.961
15	Shadhinot a Hall, PUST	24°14'12.9''N	89°53'32.1''E	0.206	0.253	0.884
16	Club Square, PUST	24° 0' 47.465" N	89° 16' 47.903" E	0.182	0.223	0.781
17	Civil Engineeri ng and Architectu re	24° 0' 45.399" N	89° 16' 48.266" E	0.202	0.248	0.867

	Laborator y					
18	Teacher's and Officer's Club, PUST	24° 0' 49.530" N	89° 16' 48.495" E	0.178	0.218	0.764
19	PUST Guest House	24° 0' 49.655" N	89° 16' 48.826" E	0.165	0.202	0.708
20	Matribhas ha Hall, PUST	24° 0' 48.458" N	89° 16' 51.384" E	0.188	0.231	0.807
21	Ganatantr a Hall, PUST	24° 0' 48.532" N	89° 16' 53.911" E	0.157	0.193	0.674
22	Bijoy 24 Hall, PUST	24° 0' 47.153" N	89° 16' 52.345" E	0.168	0.206	0.721
23	New Academic Building- 1, PUST	24° 0' 45.300" N	89° 16' 50.935" E	0.143	0.175	0.614
24	Teacher's Dormitory	24° 0' 50.818" N	89° 16' 51.309" E	0.185	0.227	0.794
25	Garden, PUST	24° 0' 41.762" N	89° 16' 45.554" E	0.221	0.271	0.949
Average				0.1826	0.22396	0.752

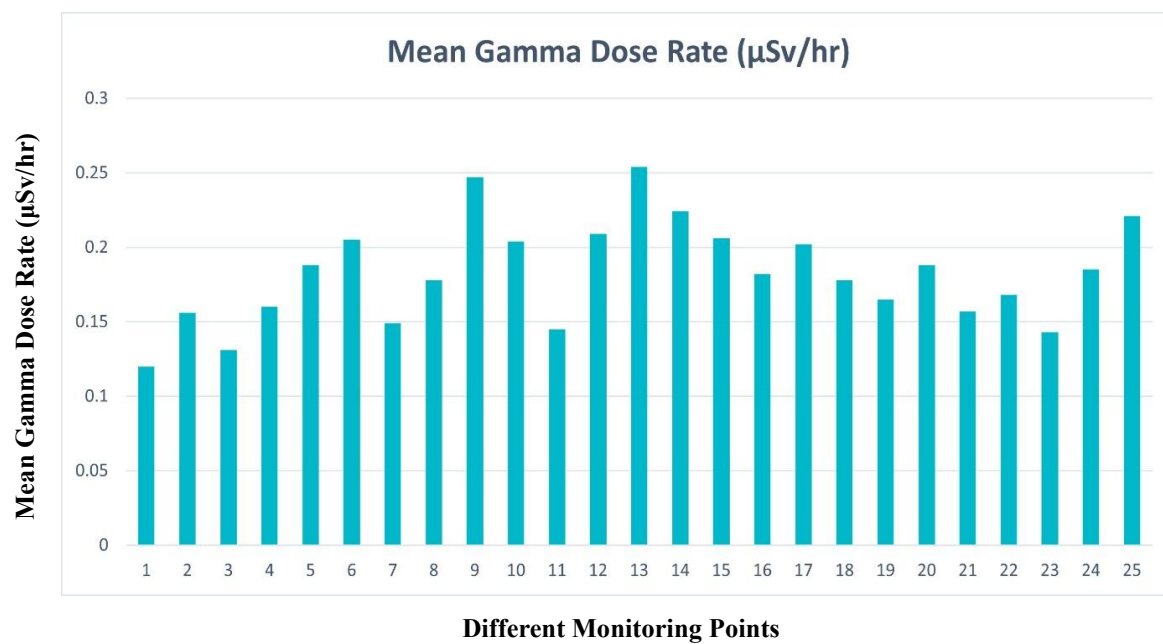
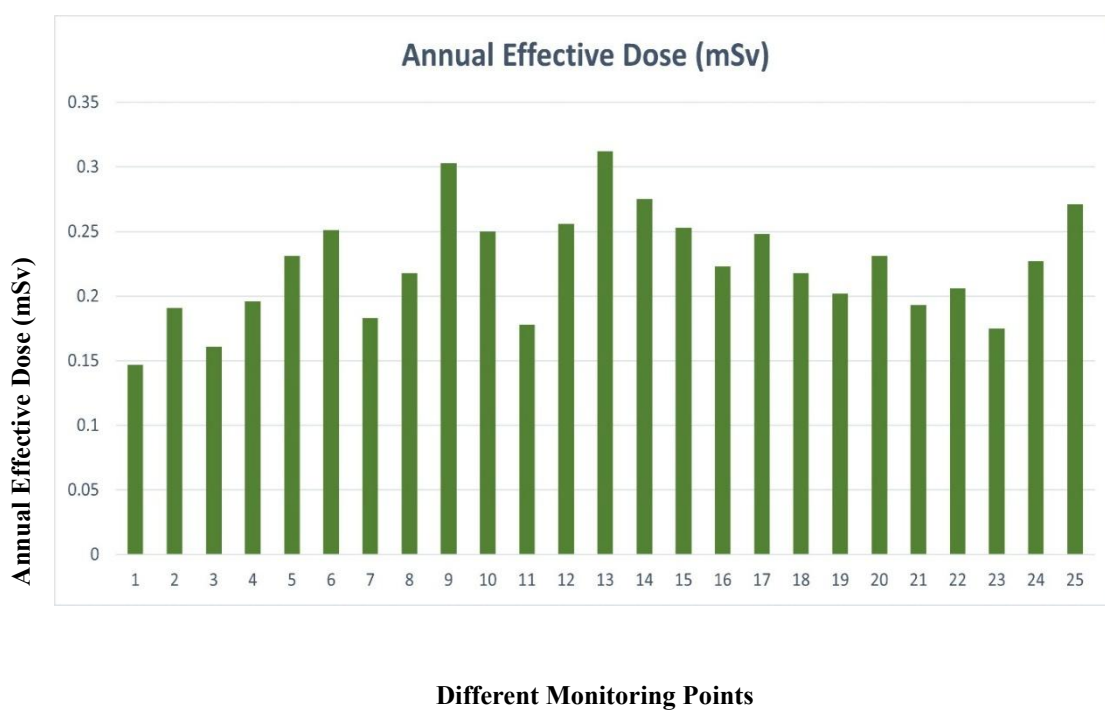


Figure 10: Variation of Mean Gamma Dose Rate in various place in PUST.



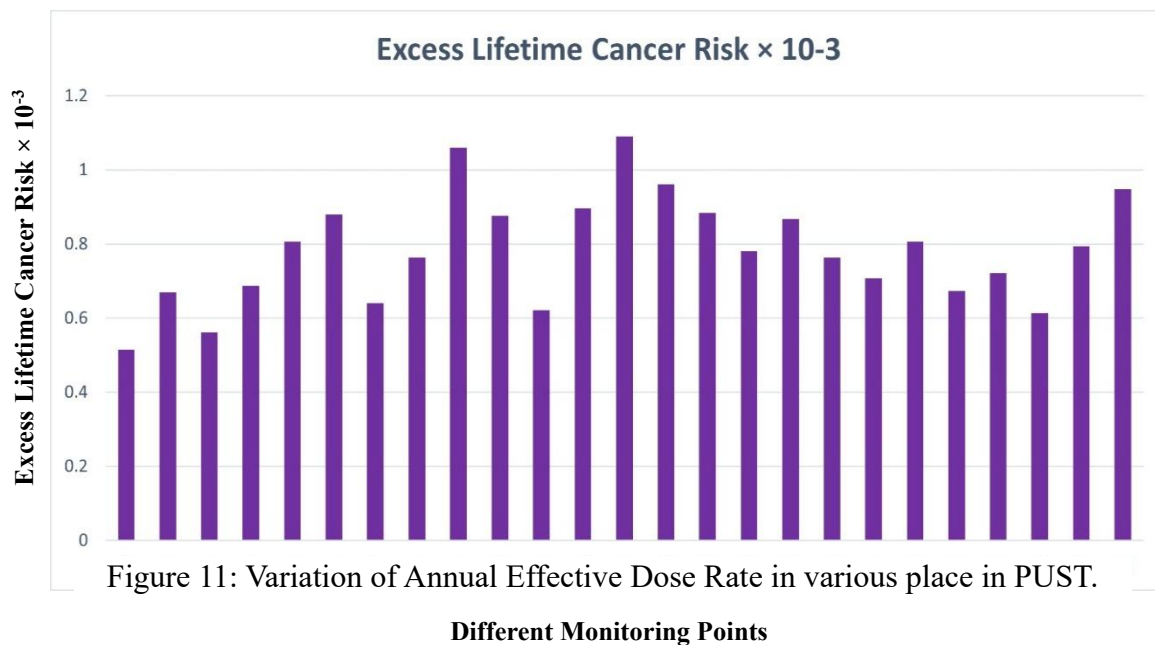
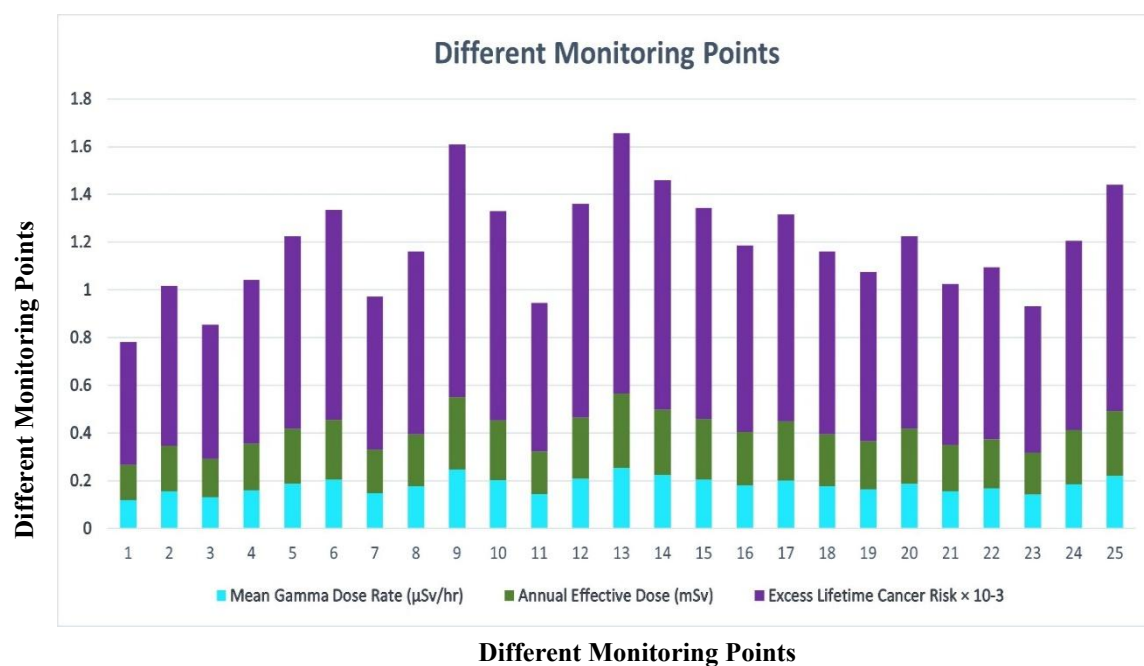


Figure 12: Variation of Excess Lifetime Cancer Risk in various place in PUST.



5.2.2 Discussion:

Figure 13 presents the variation of mean gamma dose rate, annual effective dose (AED), and excess lifetime cancer risk (ELCR) at different monitoring locations within the Pabna University of Science and Technology (PUST) campus. The measured gamma dose rates ranged from $0.120 \mu\text{Sv h}^{-1}$ at the Main Gate to $0.254 \mu\text{Sv h}^{-1}$ at the Convention Center, indicating spatial variability in the natural radiation environment of the campus. Such variations are commonly observed in environmental radiation surveys and are generally associated with differences in geological characteristics, radionuclide concentrations, soil properties, and building materials [29–32].

Higher gamma dose rates were observed at locations such as the Convention Center, Medical Center, Central Temple, Garden area, and Residence of the Vice-Chancellor. **Figure 13:** Variation of mean gamma dose rate, annual effective dose, and excess lifetime cancer risk in different monitoring points in PUST.

These elevated values may be associated with the use of concrete, bricks, and other construction materials containing naturally occurring radionuclides such as uranium, thorium, and potassium isotopes [33,34]. In addition, local soil characteristics and environmental conditions may contribute to the observed spatial distribution of radiation levels. Lower dose rates recorded at the Main Gate, Central Field, and New Academic Building-1 suggest comparatively lower concentrations of naturally occurring radioactive materials in those areas.

The AED values were found to vary between 0.147 and 0.312 mSv y^{-1} , while ELCR values ranged from 0.515×10^{-3} to 1.090×10^{-3} . As expected, both AED and ELCR followed the same trend as the measured gamma dose rate because these parameters are directly dependent on radiation exposure levels. Comparable AED and ELCR values have been reported in environmental radiation monitoring studies conducted around

university campuses, residential areas, and urban environments in Bangladesh and other countries [30,35,36].

An important aspect of this study is that the PUST campus is located comparatively close to the Rooppur Nuclear Power Plant (RNPP). Establishing baseline environmental radiation data before long-term operation of nuclear facilities is essential for future environmental impact assessment and radiological surveillance programs [39]. The present results therefore provide important reference data against which future radiation measurements can be compared. Nevertheless, the measured gamma dose rates are consistent with normal background radiation levels reported in many regions of the world and do not indicate any abnormal radiological condition [31,37].

According to UNSCEAR recommendations, variations in terrestrial gamma radiation are expected due to natural geological diversity, and moderate deviations from the world average do not necessarily imply radiological hazards [37]. Similar conclusions have been reported in studies conducted in Bangladesh, India, Cyprus, and several other countries where elevated local background radiation levels were found to originate primarily from natural sources rather than anthropogenic activities [30,32,35,38]. The calculated ELCR values remain within acceptable limits and indicate negligible radiological risk to the public. Therefore, the overall radiological condition of the PUST campus can be considered safe for students, faculty members, staff, and nearby residents. Furthermore, the data generated in this study will be useful for future environmental radiation monitoring and assessment in the vicinity of the Rooppur Nuclear Power Plant [40, 41].

5.3 Comparative Analysis: MBSTU vs PUST

Table 3: Summary of Mean Radiological Parameters — MBSTU and PUST

Parameter	MBSTU	PUST	Reference Value	PUST vs MBSTU (% diff)
GDR Range ($\mu\text{Sv/hr}$)	0.077 – 0.232	0.120 – 0.254	—	—
Mean GDR ($\mu\text{Sv/hr}$)	0.143	0.183	$\sim 0.138[5]$	+27.7%
AED Range (mSv/y)	0.094 – 0.285	0.147 – 0.312	—	—
Mean AED (mSv/y)	0.176	0.224	0.48[1] / 1.0[6]	+27.6%
ELCR Range ($\times 10^{-3}$)	0.331 – 0.996	0.515 – 1.090	—	—
Mean ELCR ($\times 10^{-3}$)	0.614	0.784	0.29[1]	+27.7%
Monitoring Points	25	25	—	—

5.3.1 Comparison of Mean Gamma Dose Rate (GDR)

The mean gamma dose rates recorded across the 25 monitoring points at each campus differed noticeably between the two institutions. At MBSTU, dose rates ranged from 0.077 $\mu\text{Sv/hr}$ at the 1st Gate to 0.232 $\mu\text{Sv/hr}$ inside the Medical Center, yielding a campus-wide mean of 0.143 $\mu\text{Sv/hr}$ (SD = 0.039 $\mu\text{Sv/hr}$). At PUST, values spanned a narrower but higher band, which from 0.120 $\mu\text{Sv/hr}$ at the Main Gate to 0.254 $\mu\text{Sv/hr}$

at the Convention Center with a campus mean of $0.183 \mu\text{Sv/hr}$ ($\text{SD} = 0.034 \mu\text{Sv/hr}$). The PUST mean GDR was therefore approximately 27.7% higher than that of MBSTU.

Within each campus, the spatial variation itself is informative. MBSTU displayed a wider absolute spread ($0.155 \mu\text{Sv/hr}$ range) compared to PUST ($0.134 \mu\text{Sv/hr}$ range), and its standard deviation was slightly larger, suggesting that localised environmental differences:

such as soil composition near the Research Pond, vegetation around the Krishnochura Lane, and construction materials at the Medical Center, which produced more pronounced point-to-point fluctuation. At PUST, readings were generally more elevated throughout the campus, with even the lowest point (Main Gate, $0.120 \mu\text{Sv/hr}$) already comparable to several mid-range MBSTU locations.

To contextualize these values, the mean outdoor gamma dose rate measured at Dhaka University campus was reported as $0.138 \mu\text{Sv/hr}$ [5]. A figure almost identical to the MBSTU mean. The PUST mean of $0.183 \mu\text{Sv/hr}$, by contrast, sits notably above this Dhaka University benchmark. Similarly, [7] documented a mean of $0.135 \mu\text{Sv/hr}$ in the New Market Thana of Dhaka. Both PUST and MBSTU means remain below the $0.20 \mu\text{Sv/hr}$ threshold that would broadly correspond to the ICRP public dose limit of 1 mSv/y when exposure is treated as continuous outdoor exposure only and both fall well within the range of normal natural background radiation observed in similar regions of Bangladesh.

The higher GDR at PUST likely reflects a combination of geological and infrastructural factors. The alluvial soils of the Pabna region, while generally of moderate radioactivity, may contain locally enriched pockets of potassium-40 and thorium-series nuclides in certain sub-surface layers. Additionally, building materials used in the newer constructions at PUST — including concrete, brick, and stone aggregate that can elevate ambient gamma dose rates relative to open or vegetated areas, particularly at indoor-adjacent monitoring points such as the Convention Center and Medical Center. The MBSTU campus, which includes more open green spaces and older, lower-density construction at its periphery, may present a somewhat lower average contribution from building materials. Notably, in both campuses the highest readings occurred in or immediately adjacent to enclosed or semi-enclosed built structures, consistent with the

well-established relationship between construction materials and elevated terrestrial gamma exposure [1].

5.3.2 Comparison of Annual Effective Dose (AED)

Since the annual effective dose is calculated directly from the measured gamma dose rate using the standard outdoor occupancy factor (0.2) and the dose conversion coefficient, the inter-campus difference in AED mirrors that of GDR precisely. The mean AED at MBSTU was 0.176 mSv/y (range: 0.094 – 0.285 mSv/y), while at PUST it was 0.224 mSv/y (range: 0.147 – 0.312 mSv/y) — again approximately 27.6% higher. Both values are comfortably below [6] recommended public exposure limit of 1 mSv/y; MBSTU's means represents only 17.6% of this limit, and PUST's represents 22.4%.

Compared to [1] world average annual effective dose from outdoor terrestrial gamma radiation, which is estimated at 0.48 mSv/y and both campuses fall well short, with MBSTU at 36.6% and PUST at 46.7% of the global mean. This indicates that neither campus environment poses a radiologically elevated external gamma burden relative to the rest of the world, though PUST's figure is closer to roughly half of the global reference. For Bangladesh specifically, the national average absorbed dose rate has been estimated at approximately 64.16 nGy/h, corresponding to an AED of around 0.313 mSv/y [2]; the MBSTU mean falls below this national average, while PUST's mean approaches but does not exceed it.

It is also instructive to look at the highest individual monitoring points at each site. The single highest AED recorded was 0.312 mSv/y at the PUST Convention Center (monitoring point 13), and the highest at MBSTU was 0.285 mSv/y at the Medical Center Inside (monitoring point 7). Even these maximum values remain below one-third of the ICRP limit, reinforcing that there are no hotspot locations within either campus that would require remedial radiological action.

5.3.3 Comparison of Excess Lifetime Cancer Risk (ELCR) and Environmental Health Implications

The excess lifetime cancer risk is perhaps the most directly interpretable health indicator arising from long-term background gamma exposure. At MBSTU, ELCR values ranged from 0.331×10^{-3} at the 1st Gate to 0.996×10^{-3} at the Medical Center Inside, with a campus mean of 0.614×10^{-3} . At PUST, values ranged from 0.515×10^{-3} at the Main Gate to 1.090×10^{-3} at the Convention Center, with a campus mean of 0.784×10^{-3} . The PUST mean ELCR was 27.7% higher than at MBSTU, consistent with the GDR and AED differences.

The world average ELCR from natural background gamma radiation is generally cited as approximately 0.29×10^{-3} [1]. Both campuses report mean ELCR values above this global reference, At MBSTU that is roughly 2.1 times and PUST at roughly 2.7 times the world average. At first glance this may seem concerning, but it requires careful interpretation. The world average figure is a population-weighted global mean that includes very low-exposure populations, and it encompasses a very wide natural range; values several times the world mean are common in ordinary terrestrial environments across Asia, including Bangladesh, where soil radioactivity tends to run higher than the global average in many districts [2]. Moreover, it is important to note that ELCR represents a theoretical upper-bound lifetime risk under continuous outdoor exposure assumptions. However, the actual individual exposure is considerably lower because people spend the majority of their time indoors, and indoor and outdoor dose rates differ.

A useful regional comparison comes from [9], who measured ELCR values exceeding 0.29×10^{-3} at the University of Port Harcourt Teaching Hospital in Nigeria even in the absence of nearby nuclear sources, demonstrating that university campus environments can routinely display elevated ELCR simply due to building materials and local geology. Studies at hospital campuses in Dhaka [8]. Similarly found ELCR values varying between locations but remaining within acceptable ranges under standard risk assessment frameworks. Critically, even the highest individual ELCR value measured in this study is 1.090×10^{-3} at PUST's Convention Center which lies below the commonly used regulatory upper-bound risk threshold of 1×10^{-3} used by some

agencies, and only marginally exceeds it; under ICRP's framework, the dose from which it is derived (0.312 mSv/y) is a small fraction of the 1 mSv/y public limit.

In summary, while PUST consistently exhibits higher gamma dose rate, annual effective dose, and ELCR values than MBSTU across all 25 monitoring points. A difference of approximately 27–28% in all three parameters. However, both campuses remain within internationally accepted radiological safety boundaries. The observed difference between the two sites most plausibly reflects the combined influence of differences in local soil mineralogy, construction material density, campus layout, and vegetation coverage, rather than any artificial radiological source. Neither site shows any indication of contamination from artificial radionuclides or radiological hazard attributable to anthropogenic radiation sources. The overall radiological environment at both PUST and MBSTU can therefore be characterized as safe for students, academic staff, administrative personnel, and the surrounding community under routine conditions of occupancy.

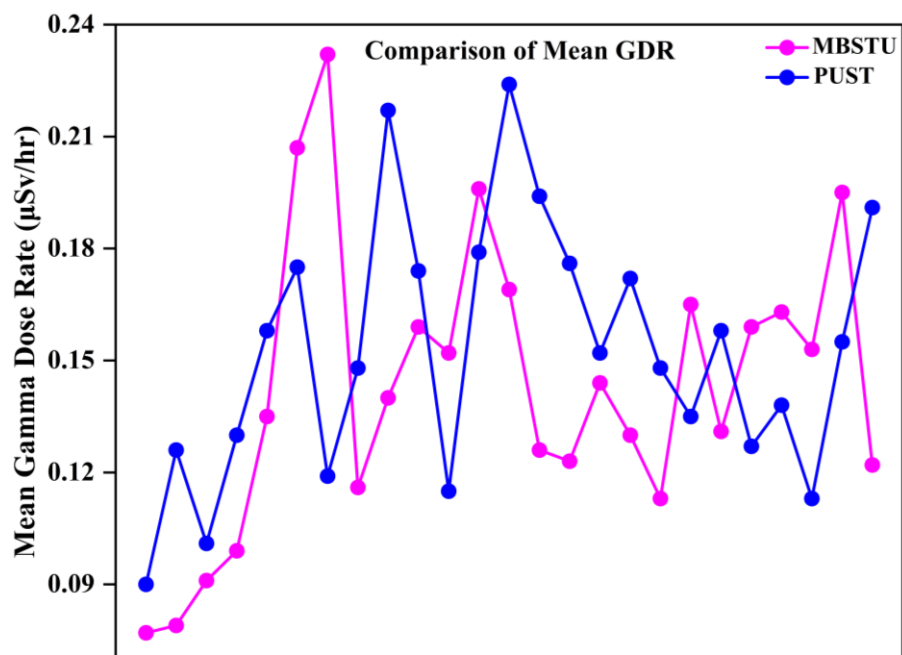


Figure 14: Comparison of mean gamma dose rate between PUST and MBSTU.

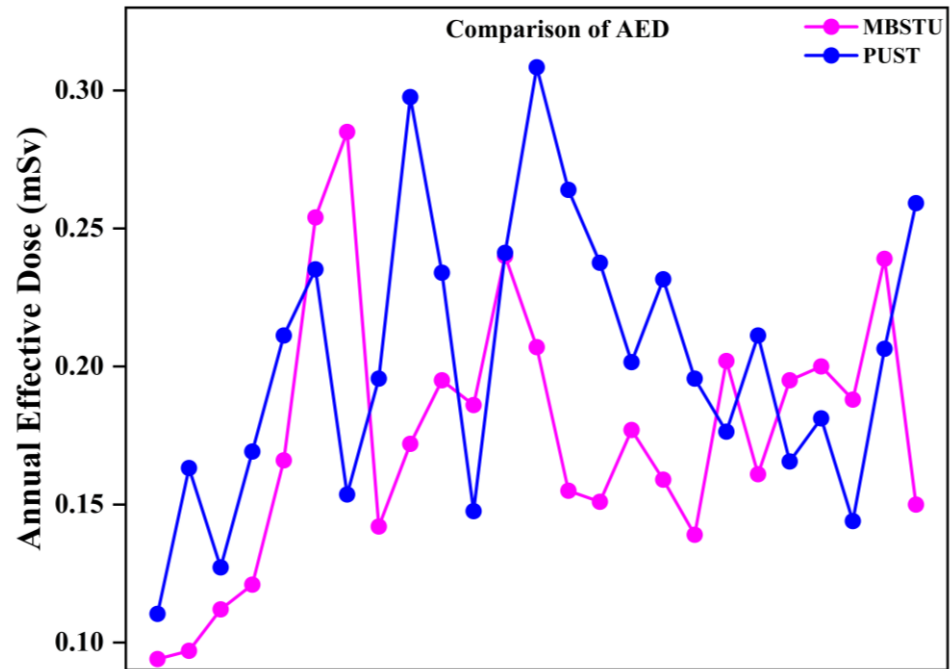


Figure 15: Comparison of annual effective dose rate between PUST and MBSTU.

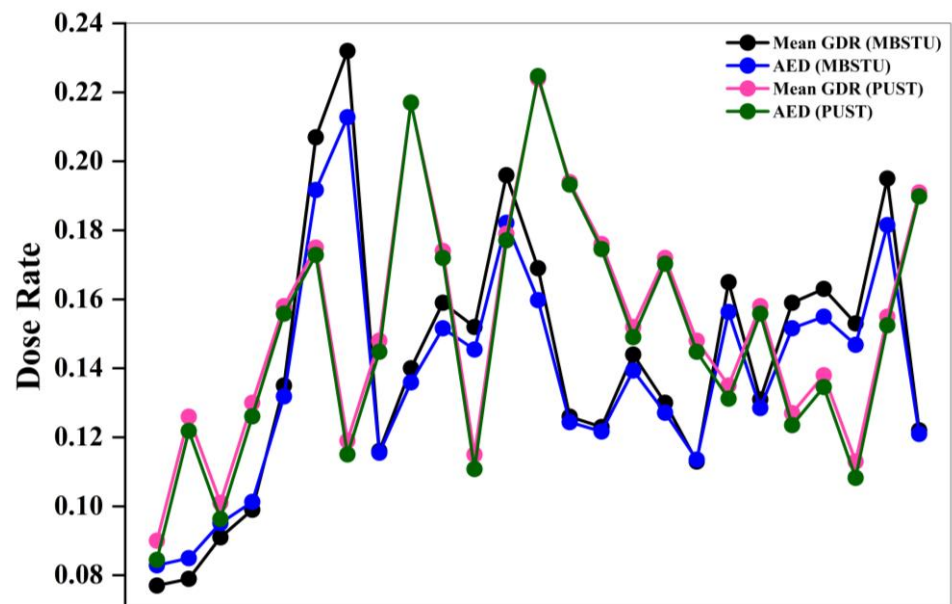


Figure 16: Comparison of gamma dose rate between PUST and MBSTU.

Chapter 6

6.1 Conclusion

The main objective of this study was to measure and compare the ambient gamma radiation level of two public universities, Mawlana Bhashani Science and Technology University (MBSTU) at Tangail and Pabna University of Science and Technology (PUST) at Pabna in Bangladesh. The calibrated GAMMA SCOUT Geiger counter was used to collect a detailed dataset for two campuses, at 25 carefully selected monitoring points per campus, of gamma dose rates, annual effective doses, and excess lifetime cancer risk. These all contribute to a few simple and understandable conclusions. The campuses are both radiations free. This is the primary lesson we can learn from the whole study. The measured gamma dose rate at MBSTU and PUST, AED and excess lifetime cancer risk (ELCR) were also found in less than the recommended gamma dose rate limit for safety level suggested by UNSCEAR and ICRP. The maximum AED during the study was 0.312 mSv/y in PUST Convention Center, which is less than one-third of the recommended public exposure dose limit (1 mSv/y) by the ICRP. Under normal, day to day conditions, the level of background gamma radiation to which students, teachers, staff and the community around both campuses are exposed is not radiologically hazardous. At all the 25 monitoring locations PUST had higher radiation readings throughout, with no exceptions. The mean gamma dose rate at PUST was 0.183 μ Sv/hr which is approximately 27.7% higher than the gamma dose rate at MBSTU (0.143 μ Sv/hr). The same was observed for the mean AED values which were 0.224 mSv/y for PUST and 0.176 mSv/y for MBSTU. Similarly, the mean ELCR at PUST (0.784×10^{-3}) was about 27.7% higher than at MBSTU (0.614×10^{-3}). There are statistically significant and consistent differences, but not at a level of safety. It is a natural difference and not a nuclear difference. The nearness of PUST to the construction site of the Rooppur Nuclear Power Plant (RNPP), located at Ishwardi District, 31 km away, where fuel installation started in early 2026, was one of the key concerns in this study. The high dose rates at PUST however have no signature of artificial radionuclide contamination. The difference between the two campuses is more

because of natural factors - variation in mineralogy of the local soils, density and type of building materials used at newer buildings constructed at PUST and MBSTU, campus layout and relative absence of open green buffer zones at PUST compared to MBSTU. The increased readings at closed or partially closed sites, like the PUST Convention Center and Medical Center, are in complete alignment with the very well documented phenomenon that concrete, brick and stone aggregates can elevate ambient gamma levels as compared to open outdoor areas. The PUST campus level does not show any radiological influence whatsoever due to the RNPP. There is spatial variation within each campus and its own story; the lowest reading at MBSTU was at the 1st Gate ($0.077 \mu\text{Sv/hr}$) and the highest reading was in the Medical Center ($0.232 \mu\text{Sv/hr}$). The range of the PUST was from the Main Gate ($0.120 \mu\text{Sv/hr}$) to the Convention Center ($0.254 \mu\text{Sv/hr}$). The differences within the campus are due to microenvironmental variations such as soil type adjacent to water bodies, vegetation density, age of buildings and building materials. No 'hotspot' areas were found at either site that posed a health risk and needed remediation. This study is useful in filling up the gap in the radiological database of Bangladesh. Before this study, there were no published in-situ baseline gamma radiation data either at MBSTU or PUST. Having baseline measurements at a nearby campus such as PUST, ahead of the commissioning of the RNPP, is not only academically valuable, but is necessary. Pré-operational baseline measurements at a nearby campus like PUST, prior to the commissioning of the RNPP, are not only academically useful, but are practically essential. The data are baseline data from which comparisons can be made with any future post-operational measurements. If the radiation level at PUST should ever shift in the future, there will be a baseline of radiation levels to compare with. Finally, the authors demonstrated the feasibility and benefit of frequent radiation monitoring on the college campus. It does not require extensive infrastructure but instead can be achieved using an appropriately calibrated portable detector and protocol in the field to generate policy-relevant, credible data. It was observed that both MBSTU and PUST were radiologically safe environment, and it can be assured with reference to internationally accepted standards. This work is hoped to be a good inspiration for other University based radiological baseline surveys across the country to develop a more complete national radiological monitoring network to guide Bangladesh's future for development of nuclear energy.

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